

# Wangjie Su

Undergraduate | College of Engineering | Peking University

+86 188 6803 2449 | [wsu0605@stu.pku.edu.cn](mailto:wsu0605@stu.pku.edu.cn)

<https://github.com/csxd0605> | <https://csxd0605.github.io/>

## Research Interests

**Current focus:** Reliable AI agents for scientific simulation and industrial workflows, with an emphasis on domain-tool and knowledge integration, long-horizon memory and context, multi-agent orchestration, failure recovery, and evaluation.

**Future directions:** Grounding long-horizon ReAct agents in multimodal perception, sensor streams, world models, and physical action to build perception–planning–action–feedback loops for autonomous laboratories, scientific instruments, and general-purpose robots.

## Education

Peking University, B.S. in Theoretical and Applied Mechanics

Sep 2023 – Present

Peking University, Double Degree in Intelligent Science and Technology

Sep 2024 – Present

- **GPA:** 3.818/4.0
- **Selected Coursework:** Data Structures and Algorithms (**100**), Programming in AI (**100**), Computer Vision, Probability and Statistics, Numerical Analysis, Generative Modeling, Intelligent Robotics, Trustworthy ML, LLMs and NLG, and Physics Simulation for Computer Graphics.

## Research Experience

**AI + Fusion Energy Series Project**

Undergraduate Researcher

Mar 2026 – Present

Xiniao Technology Innovation Center, Peking University | PIs: Prof. Yunfeng Liang and Prof. Songfang Huang

- **Problem:** Developing AI agents that can carry out long-horizon fusion-energy simulation workflows, from task planning and knowledge retrieval to tool execution, result interpretation, and iterative refinement.
- **Implementation:** Using **Nexgent** as the experimental harness to connect domain knowledge and simulation tools through MCP, coordinate specialist agents and explicit workflows, and preserve sessions, checkpoints, permissions, and tool traces.
- **Research focus:** Investigating long-context retrieval and recovery, sustained adherence and progress in long-horizon tasks, and permission-gated failure recovery so scientific runs remain state-consistent, auditable, and reproducible.

**Combustion Laboratory, Peking University**

Undergraduate Researcher

May 2025 – Present

PI: Prof. Zheng Chen

- **Background:** Studying fundamental combustion problems in energy conversion and propulsion, including flame dynamics, chemical kinetics, transport, ignition, flame propagation and stabilization, and laminar flame speed.
- **Current work:** Exploring mathematical formulations of multiscale combustion processes from conservation laws and reaction mechanisms, together with machine-learning approaches to predicting key quantities and approximating complex processes.
- **Methodological focus:** Comparing the regimes of mechanistic and data-driven models, with attention to physical consistency, interpretability, data efficiency, and out-of-distribution generalization toward physics–data integrated combustion modeling.

## Work Experience

**Hangzhou Yuanxi Technology Co., Ltd.**

Independent Project Lead

Jul 2026 – Present

Early Technical Team Member | System: ManySelves — An Independently Designed General-Purpose Agent Capability Substrate

- **Capability definition:** Reframed an agent team as a versionable capability specification rather than a fixed application or prompt bundle: Identity contracts define roles, boundaries, I/O, and handoffs; Skill documents encode reusable methods and acceptance criteria; a stable runtime realizes both without domain-specific rewrites.

- **Execution abstraction:** Independently built the system end to end around one Main Agent and task-scoped specialists operating on project-local knowledge, state, and artifacts. Typed state, checkpoints, rollback, resumable decisions, and explicit review boundaries make long-horizon behavior inspectable and recoverable.
- **Research platform:** FastAPI exposes only conversation and task submission while model/tool orchestration, state evolution, and artifact provenance remain server-side. This creates a controlled platform for studying capability transfer, multi-agent coordination, memory, and recovery, with reuse across different task workflows and future tasks.

## Selected Projects

---

**Nexgent: Scientific Agent Harness** *Independent Project* **May 2026 – Present**

- **Problem:** Long-running scientific tasks need more than a chat interface: agents must connect to domain tools, coordinate specialized work, survive interruptions, expose risky actions, and leave a traceable experimental record. Nexgent was developed as the Agent Harness for prototyping these requirements in the AI + Fusion Energy project.
- **System:** Built a model-agnostic, frontend-neutral runtime shared by a PyQt6 desktop workspace, Textual TUI, and headless CLI; integrated 38 tools, automatic MCP tool bridging, multi-model routing, and sub-agent/workflow orchestration in pipeline, parallel, and phased modes.
- **Workflow functions:** Implemented progressive context compression, typed project/reference memory, session save–resume–fork, checkpoint rollback, permission gates, lifecycle hooks, goals, and background tasks for controlled and recoverable experiment execution.

**oh-my-opencode (now oh-my-openagent)** *Open-Source Contributor* **May 2026 – Present**

- **Research question:** Investigated which agent behaviors should remain invariant across OpenCode, Codex, Pi, and Claude Code rather than being entangled with one Harness. Proposed a downward-only Core–Adapter DAG separating reusable policy from Hook, provider, and UI effects.
- **Semantic abstractions:** Defined agent configuration as a zero-dependency core and model fallback as a session-scoped idle/armed/exhausted protocol with injected reachability and logging, turning implicit error handling into explicit, portable, and testable state transitions.
- **System validation:** Preserved existing call sites through re-export shims and tested cleanup, re-arming, chain exhaustion, and session isolation; continue to help lead the monorepo refactor, putting reusable Harness-independent policies into practice through plugins.

**Where Is My Codex (Flow Rescue)** *Course Project* **Spring 2026**

- **Physics:** Co-developed a 2D fluid-puzzle game and implemented Position-Based Fluids in Python/Taichi with uniform-grid neighbor search, density constraints, terrain collisions, and GPU/CPU backends.
- **Engineering:** Built five playable levels and a Tiled import pipeline; added headless tests and profiling for digging, collision, channel stability, goal collection, grid construction, and constraint solving.

**Tiny PyTorch** *Course Project* **Fall 2024**

- **Framework:** Built a PyTorch-style tensor and reverse-topological autodiff engine with core operators, fully connected layers, convolution, pooling, softmax loss, and end-to-end MLP/CNN training.
- **Acceleration:** Implemented C++/CUDA storage, transfers, custom kernels, and cuBLAS-backed computation with Python bindings; trained on MNIST and compared single-GPU and DataParallel workflows.

## Honors & Awards

- 
- National Scholarship and Merit Student, Peking University (2025)
  - Second Prize, Beijing Division, National College Student Mathematical Modeling Competition (2025)
  - Outstanding Learning Award and Third Scholarship, Peking University (2024)
  - First Prize, Beijing College Student Mathematics Competition (2024)

## Skills

---

**Programming:** Python, C/C++, CUDA, MATLAB, LaTeX, CMake, Taichi

**Tools & Frameworks:** PyTorch, PyQt6, Git, Linux, Scikit-learn, Jupyter, OpenFOAM

**Languages:** Chinese (native); English (CET-6: 613)



Name: Su Wangjie  
Department: College of Engineering  
Major: Theoretical and Applied Mechanics

Sex: Male      DOB: Jun.05,2005  
Period of Study: 2023.09-2027.06  
ID: 2300011075

| Academic year | Course Title                                                                               | Hours | Unit | Grade | Academic year | Course Title                                                                          | Hours | Unit | Grade |
|---------------|--------------------------------------------------------------------------------------------|-------|------|-------|---------------|---------------------------------------------------------------------------------------|-------|------|-------|
| 23-24/1       | Introduction to Psychology                                                                 | 32    | 2    | 85    | 24-25/2       | Critical Thinking and Academic WritingΔ                                               | 34    | 2    | 88    |
|               | Linear Algebra and Geometry                                                                | 85    | 4    | 89    |               | Circuits and ElectronicsΔ                                                             | 54    | 3    | 88    |
|               | Social practice and service learning, Part I                                               | 32    | 1    | P     |               | Computer Vision+                                                                      | 51    | 3    | 96    |
|               | Mathematical Analysis(I)                                                                   | 96    | 4    | 90    |               | Ordinary Differential Equations                                                       | 68    | 3    | 94    |
|               | Engineering Drawing                                                                        | 54    | 3    | 90    |               | Probability and Mathematical Statistics                                               | 51    | 3    | 96    |
|               | Events and Policies                                                                        | 34    | 2    | P     |               | Numerical Analysis                                                                    | 85    | 3    | 93    |
|               | Introduction to Computation(B)                                                             | 48    | 3    | 100   |               | Fencing                                                                               | 32    | 1    | 94    |
|               | Swimming                                                                                   | 32    | 1    | 89    |               | Engineering Thermodynamics                                                            | 51    | 3    | 96    |
|               | Military Theory                                                                            | 32    | 2    | 93    |               | The Basic Principles of Marxism                                                       | 51    | 3    | 85    |
|               | Introduction to Modern Engineering                                                         | 34    | 1    | P     |               | Mathematics in Engineering                                                            | 68    | 3    | 89    |
|               | Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era | 51    | 3    | 90    |               | Jewish Civilization in a Global Context                                               | 34    | 2    | P     |
|               | College Chemistry                                                                          | 32    | 2    | 81    |               | Basic Physics Lab                                                                     | 68    | 2    | 86    |
| 23-24/2       | General Chemistry(B)                                                                       | 64    | 4    | 92    |               | Engineering Fluid Mechanics                                                           | 51    | 3    | 96    |
|               | General Physics(1)                                                                         | 68    | 4    | 84    | 24-25/3       | Advanced Dynamics                                                                     | 51    | 3    | 92    |
|               | Mathematical Analysis(II)                                                                  | 96    | 4    | 84    |               | Mechanics of Materials B                                                              | 68    | 3    | 88    |
|               | Social practice and service learning, Part II                                              | 32    | 1    | P     |               | Fundamentals of Control TheoryΔ                                                       | 45    | 3    | 96    |
|               | Politics and Diplomacy of Modern China                                                     | 32    | 2    | 84.5  |               | A Survey of Mao Tsetung Thoughts and Theory of Socialism with Chinese Characteristics | 51    | 3    | 94    |
|               | Outline of Chinese Modern History                                                          | 48    | 3    | 84    |               | Introduction to Intelligent Robots                                                    | 51    | 3    | 90    |
|               | Data Structure and Algorithm(B)                                                            | 48    | 3    | 100   |               | Structural Mechanics of Flight Vehicles                                               | 51    | 3    | 92    |
| 24-25/1       | Advanced Algebra(II)                                                                       | 80    | 4    | W     | 25-26/1       | Ideology, morality and Law                                                            | 51    | 3    | 92    |
|               | Programming in Artificial Intelligence+                                                    | 51    | 3    | 100   |               | Introduction to Generative Modeling+                                                  | 51    | 3    | A+    |
|               | General Physics(2)                                                                         | 68    | 4    | 93.5  |               | Aircraft Design and Propulsion                                                        | 51    | 3    | 94    |
|               | Theoretical Mechanics                                                                      | 68    | 3    | 90    |               |                                                                                       |       |      |       |

Registrar:

Educational Administration Division



Note: Courses marked with \* indicate credits transferred from other universities including overseas academic institutions (Unit and Grade are original). Courses marked with Δ indicate English taught courses at Peking University. Courses marked with + indicate minor or double major program courses. Courses marked with # indicate retaken courses. Courses marked with [G] are graduate-level courses taken during the undergraduate phase by students enrolled in the integrated undergraduate-to-graduate training program. Other Designations in Grading: P(Pass), NP(Not Pass), I(Incomplete), IP(In Progress), W(Withdraw), F(Fail), EX(Exempt).

Printing Date: Jul.14,2026



Name: Su Wangjie  
Department: College of Engineering  
Major: Theoretical and Applied Mechanics

Sex: Male      DOB: Jun.05,2005  
Period of Study: 2023.09-2027.06  
ID: 2300011075

| Academic year | Course Title                      | Hours | Unit | Grade | Academic year | Course Title | Hours | Unit | Grade |
|---------------|-----------------------------------|-------|------|-------|---------------|--------------|-------|------|-------|
| 25-26/2       | Analytical WritingΔ               | 34    | 2    | 93    |               |              |       |      |       |
|               | Art and aesthetic                 | 34    | 2    | P     |               |              |       |      |       |
|               | Machine Learning                  | 51    | 3    | W     |               |              |       |      |       |
|               | Trustworthy Machine Learning+     | 34    | 2    | 99    |               |              |       |      |       |
|               | Computational Fluid Dynamics      | 51    | 3    | 92    |               |              |       |      |       |
|               | Shadowboxing                      | 32    | 1    | 93    |               |              |       |      |       |
|               | Aerodynamic Foundation            | 68    | 4    | 91    |               |              |       |      |       |
|               | Experiments in Mechanics of       | 34    | 1    | 84    |               |              |       |      |       |
|               | Materials                         |       |      |       |               |              |       |      |       |
|               | Nuclear Science and Human         | 34    | 2    | 96    |               |              |       |      |       |
|               | Subsistence and Development       |       |      |       |               |              |       |      |       |
|               | Large Language Models and Natural | 34    | 2    | P     |               |              |       |      |       |
|               | Language Generation               |       |      |       |               |              |       |      |       |
|               | -----END-----                     |       |      |       |               |              |       |      |       |

Registrar:

Educational Administration Division



Note: Courses marked with \* indicate credits transferred from other universities including overseas academic institutions (Unit and Grade are original). Courses marked with Δ indicate English taught courses at Peking University. Courses marked with + indicate minor or double major program courses. Courses marked with # indicate retaken courses. Courses marked with [G] are graduate-level courses taken during the undergraduate phase by students enrolled in the integrated undergraduate-to-graduate training program. Other Designations in Grading: P(Pass), NP(Not Pass), I(Incomplete), IP(In Progress), W(Withdraw), F(Fail), EX(Exempt).

Printing Date: Jul.14,2026